

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
Higher 2

CANDIDATE  
NAME

CLASS

TUTOR'S  
NAME

## BIOLOGY

**9747/02**

Paper 2 Longer Structured Questions

**11 September 2008**

**2 hours**

Candidates answer on the Question Paper.

No Additional Materials are required

### READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

#### Section B

Answer **either** question 8 **or** 9.

You are advised to spend about one hour on each section.

At the end of the examination, fasten all your work securely together.

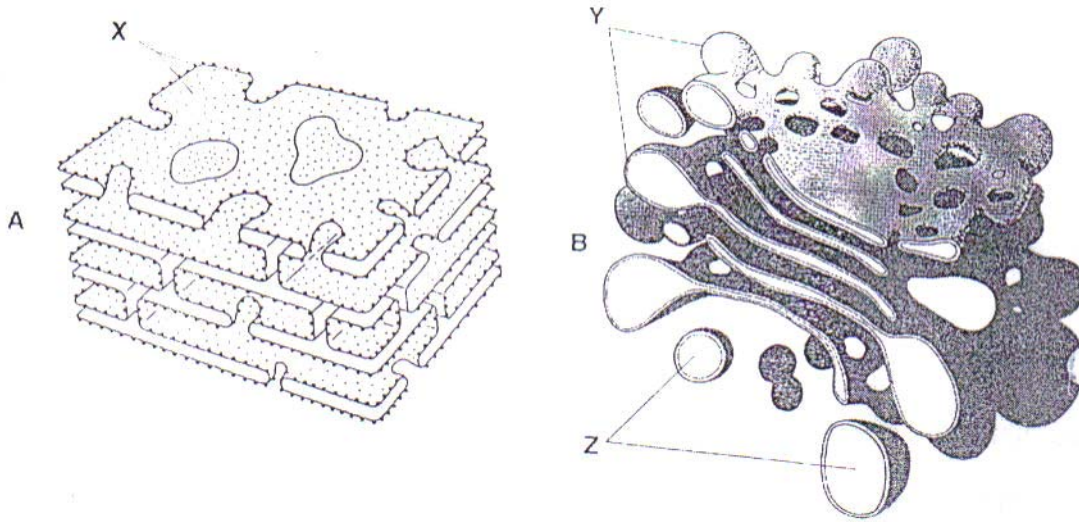
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| 1                  |  |
| 2                  |  |
| 3                  |  |
| 4                  |  |
| 5                  |  |
| 6                  |  |
| 7                  |  |
| 8 OR 9             |  |
| Total              |  |

**Section A**

Answer **all** the questions in this section

1. **Figure 1.1** below shows 2 organelles from an animal cell.



**Fig 1.1**

- (a) (i) Name the Organelles A and B [1]

A:

\_\_\_\_\_

B:

\_\_\_\_\_

- (ii) Underline the cells in the following list that do not have any A. [1]

liver cell; bacterium; virus; muscle fibre

- (iii) Name the structures Y and Z shown in Fig. B. [1]

Y: \_\_\_\_\_

Z: \_\_\_\_\_

(iii) Explain the function of A and B with relation to protein synthesis [4]

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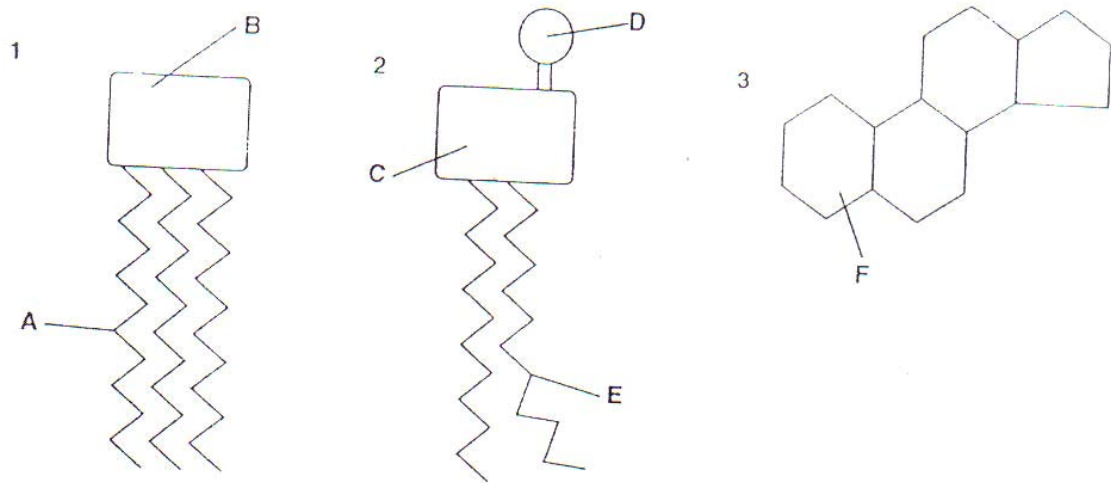
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**Figure 1.2** below shows diagrams that represent certain biomolecules found in an animal cell.



**Fig 1.2**

(b) (i) Which are the three most common elements in these molecules? [1]

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(ii) Give the number of the molecules (1, 2 or 3) which contains a fourth element and name this additional element. [1]

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- (c) (i) What general name is given to molecules with a structure similar to diagram 3? [1]

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(ii) Suggest one important function in mammals for this type of molecule. [1]

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(iii) Explain how the structure of molecule 2 is related to its function in the cell membranes. [3]

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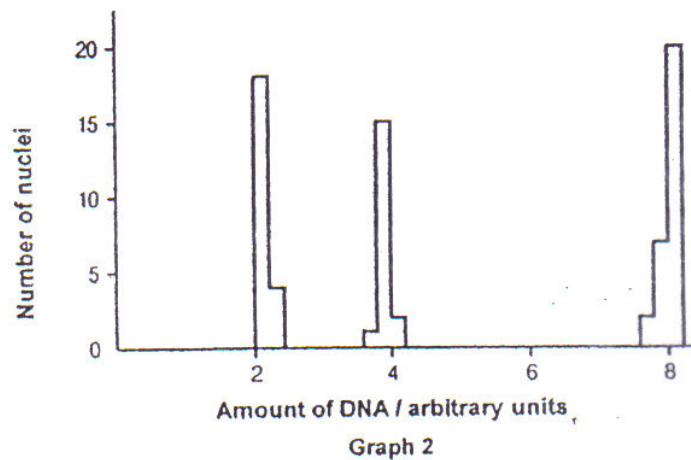
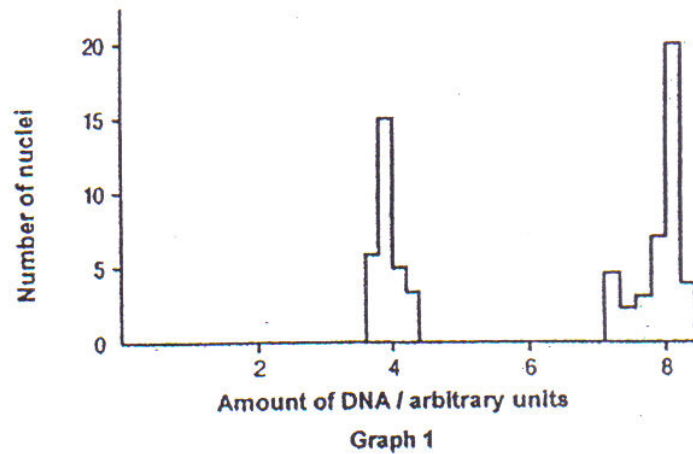
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**[Total: 14 marks]**

2. The formation of sperm cells involves mitosis and meiosis. The graphs below show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis.



With reference to the graphs,

(a) (i) Explain the evidence which shows that Graph 1 represents cells undergoing mitosis. [1]

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(ii) Explain how the events of meiosis result in three groups of nuclei in Graph 2. [2]

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(b) Explain how meiosis can produce daughter cells with slightly different genetic makeup. [2]

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(c) Cancer is a disease caused by altered gene expression. Describe how an abnormal p53 gene can give rise to a cancerous cell. [3]

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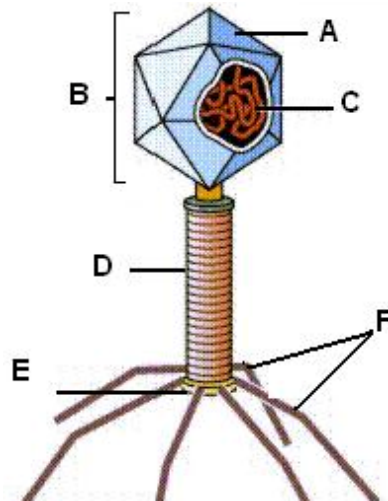
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**[Total: 8 marks]**

3. **Figure 3.1** shows a T4 bacteriophage.



**Fig 3.1**

(a) (i) Explain what is meant by the term bacteriophage [1]

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(ii) Identify the structures labeled A to F [3]

|       |    |          |
|-------|----|----------|
| A:    | D: |          |
| _____ |    | _____    |
|       | B: | _____ E: |
| _____ |    | _____    |
| C:    | F: | _____    |
| _____ |    | _____    |

(b) The Human Immunodeficiency Virus (HIV) is a retrovirus. Outline the main events of what happens after a HIV virus enters host cell. [4]

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Figure 3.2 shows a TTP molecule.

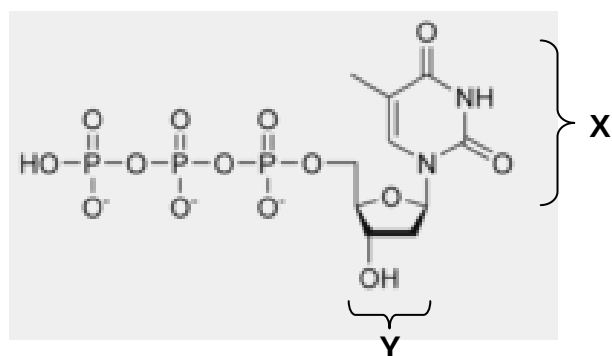


Fig 3.2

(c) (i) Name the compounds labeled X and Y.[1]

X \_\_\_\_\_  
Y \_\_\_\_\_

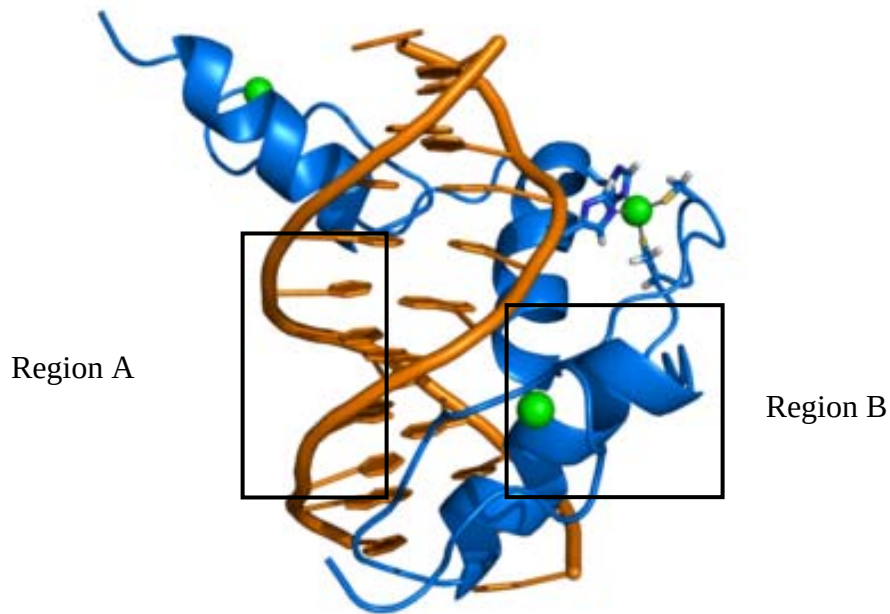
Azidothymidine (AZT), a nucleoside analogue of TTP, is used as an anti-retroviral medication to treat AIDS. The normal 3' OH group of the sugar is replaced by a azido functional group. It works as a competitive inhibitor to reverse transcriptase. Incorporation of this analogue results in chain termination.

(d) Suggest a mechanism of how this drug works. [2]

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

[Total: 11 marks]

4. **Figure 4.1** illustrates the binding of a protein with a DNA molecule.



**Fig 4.1** Zinc finger DNA complex

(a) With reference to **Fig 4.1**,

- (i) Describe in what ways the helical structure shown in Region A and B differ. [3]

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Region B contains a zinc finger protein motif. It is a highly conserved protein motif and the binding specificity is conferred by a short stretch within the  $\alpha$ -helix domain. Transcription factors often contain zinc finger motifs.

- (ii) Suggest why a transcription factor only binds to certain parts of a DNA molecule. [3]

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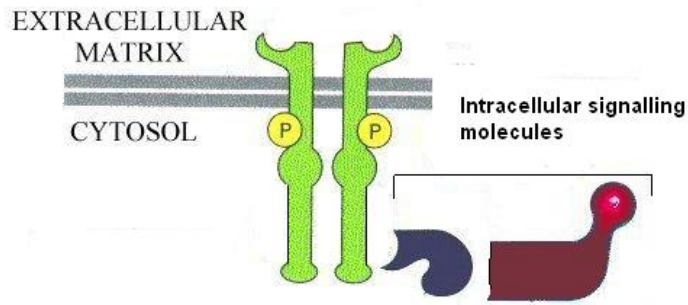


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**Figure 4.2** shows a type of cell receptor molecule.



**Fig 4.2**

(b) (i) State the three main stages of cell signaling. [1]

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(c) With reference to **Fig 4.2**,

(i) State the type of receptor shown in the diagram. [1]

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(ii) Explain how it transmits information from the extracellular to the intracellular environment [2]

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(d) Ras is a G-protein involved in signal transduction. Ras is also known as a proto-oncogene. When mutated, it often is permanently activated resulting in cancer.

(i) Explain what is signal transduction. [2]

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**[Total: 12 marks]**

5. **Figure 5.1** shows a diagram of a telomere. Telomeres possess a unique single stranded G-rich 3' overhang that loops back to form a hairpin structure as shown.

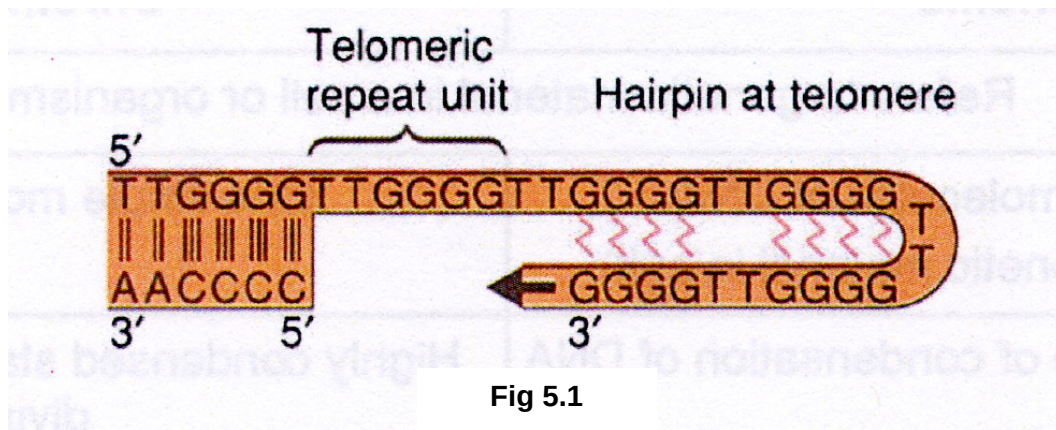


Fig 5.1

- (a) Describe the role of telomeres [3]

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- (b) With reference to **Fig 5.1**,

- (i) Describe the term tandem repeats [1]

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- (c) The telomerase enzyme is responsible for extending the telomere. It is essential in single celled organisms, while in humans and other vertebrates, it is active in reproductive cells.

- (i) Explain how the 3' overhang is formed at the end of a linear chromosome [3]

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- (ii) Explain why regeneration of telomeres is not normally required in human body cells. [2]

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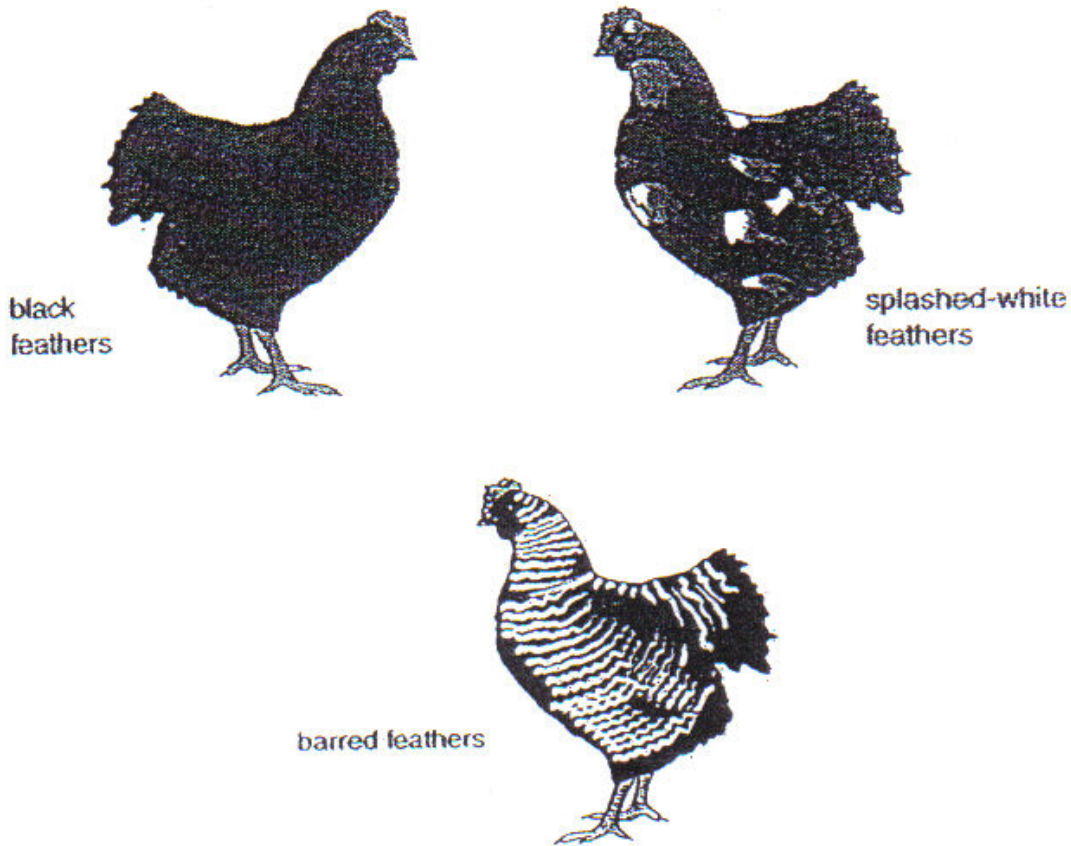
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[Total: 9 marks]

6. A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black ( $C^B$ ) and splashed-white ( $C^W$ ). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.



Another gene may cause stripes on feathers (barred feathers). The gene is carried on the X chromosome. The allele for barred feathers ( $X^A$ ) is dominant to the allele for non-barred feathers ( $X^a$ ). In chickens the male is homogametic, while the female is heterogametic.

- (a) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

- (i) Using the symbols given on the previous page draw a genetic diagram to show this cross.  
[4]

| Parental phenotype  | Male, black, non-barred feathers | Female, splashed white, barred feathers |
|---------------------|----------------------------------|-----------------------------------------|
| Genotype            |                                  |                                         |
| Gametes             |                                  |                                         |
| Offspring genotypes |                                  |                                         |
| Phenotypes          |                                  |                                         |

- (ii) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers. [3]

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- (b) The table below shows the genotypes and wool length in two breeds of sheep, X and Y, and their F1 hybrids.

|                        | Genotype | Mean wool length/cm |
|------------------------|----------|---------------------|
| Breed X                | DDEEff   | 18                  |
| Breed Y                | ddeeFF   | 14                  |
| F1 (Breed X x Breed Y) | DdEeFf   | 16                  |

When the F1 were crossed among them, the F2 offspring had a mean wool length ranging from 10 cm to 22 cm.

Assume that the inheritance of wool length in sheep depends upon alleles at three loci acting additively and that all variation in wool length in the F2 population is due to the segregation of alleles at these three loci.

What proportion of the F2 offspring are expected to have a mean wool length of 22 cm? Explain your reasoning. [4]

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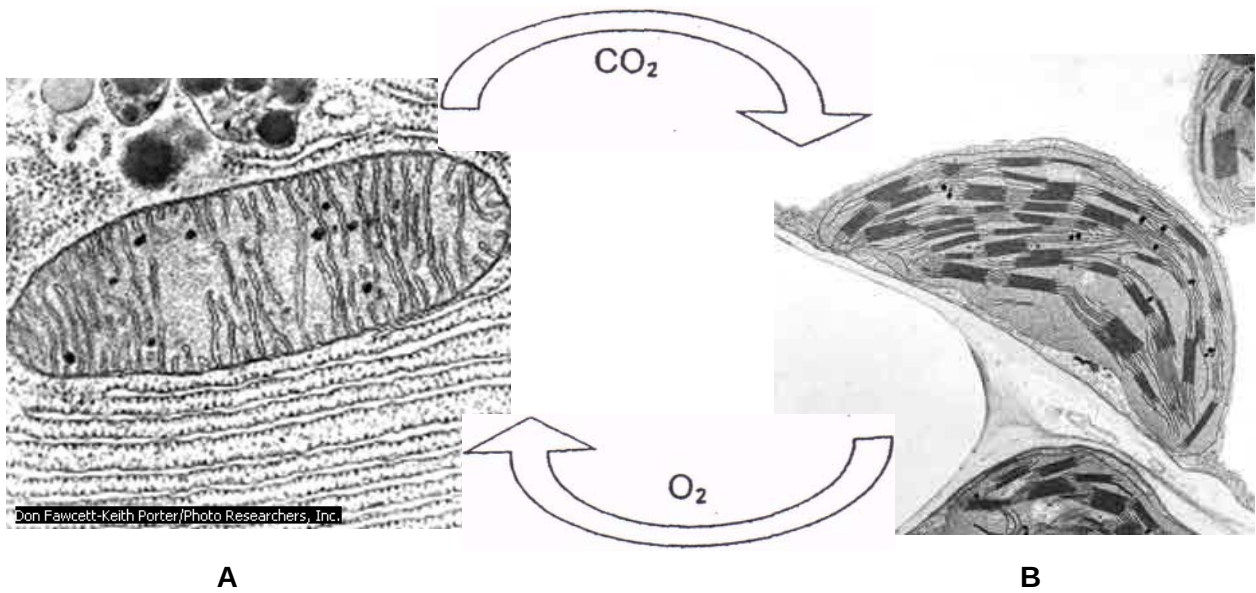
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**[Total: 11 marks]**

7. **Figure 7.1** below shows the movement of  $O_2$  and  $CO_2$  between two organelles, A and B, in a mesophyll cell of a leaf, in the presence of light



**Fig 7.1 Movement of  $O_2$  and  $CO_2$  between organelles A and B**

(a) State the process taking place in:

(i) organelle A:

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(ii) organelle B: [1]

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(b) Briefly describe how organelle B is structurally adapted to carry out its function(s). [2]

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- (c) ATP synthesis takes place in both organelles A and B. State three ways in which the process of ATP synthesis differs in the two organelles. [3]

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- (d) Briefly describe the reaction/s that bring/s about the release of CO<sub>2</sub> in organelle A. [3]

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- (e) In humans, certain tissues can undergo anaerobic respiration if conditions make it necessary.

- (i) Explain why large stores of glycogen enable these tissues to tolerate anaerobic conditions for longer period of time. [3]

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- (ii) During anaerobic respiration, pyruvate is converted to lactate. Explain the significance of this conversion. [3]

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**[Total: 15 marks]**

**Section B**

Answer **either** question 8 **or** 9 in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8. (a) Explain how the structure of two named cellular organelles are adapted for genetic expression. [8]
- (b) Explain how the changes at the genome level can lead cancer. [7]
- (c) Discuss, in terms of prokaryote and eukaryote genomes, the structure and organization of the mitochondria genome. [5]
9. (a) Explain Neo-Darwinism and compare/contrast Neo-Darwinism with the Neutral Theory. [11]
- (b) Discuss the advantages of molecular methods in tracing of human ancestry using mitochondrial DNA. [5]
- (c) Suggest reasons why Human is believed to have evolved from apes. [4]